ERGO-AI DESK: A MULTIDOMAIN ERGONOMIC RISK ASSESSMENT SYSTEM INTEGRATING COMPUTER VISION-BASED POSE ESTIMATION AND PSYCHOLOGICAL SELF-REPORT INSTRUMENTS

Waktu	Keterangan	Visual	Narasi (english)
0:00 - 0:20	Hook	Timelapse seorang mahasiswa/pekerja di depan laptop; posturnya perlahan membungkuk. Close-up tangan memijat leher.	”How do we know whether our working environment is truly ergonomic? Office workers spend hours in front of computers, often maintaining static postures that may lead to musculoskeletal discomfort and psychological strain. Yet, ergonomic risks are not always visible at first glance.”
0:20 - 0:45	Masalah	Motion graphic peta dunia dan ikon tubuh dengan titik nyeri (leher, bahu, punggung bawah) menyala merah.	”Poor working posture, prolonged static activities, and suboptimal workstation design can contribute to physical ergonomic risks. At the same time, psychological factors, including perceived stress, may further influence workers’ overall well-being. These interconnected dimensions highlight the need for a more comprehensive approach to ergonomic risk assessment.”
0:45 - 1:00	Celah Penelitian	Layar terbagi dua: kiri, observer mengisi lembar RULA di clipboard; kanan, kuesioner kertas. Garis putus-putus memisahkan keduanya, lalu muncul tanda tanya di tengah.	”Conventional ergonomic assessments often evaluate physical posture and psychological conditions separately. Physical assessments may rely on manual observation, while psychological conditions are measured through self-report questionnaires. When these assessments are not integrated, the interaction between physical and psychological risks may be overlooked.”

1:00 - 1:40	Solusi	ERGO AI DESK muncul. Animasi diagram alur: Kamera → Pose Estimation → Skor Fisik; Kuesioner → Skor Psikologis; keduanya menyatu menjadi Indeks Risiko Multidomain.	“We developed ERGO-AI DESK, an AI-assisted multidomain ergonomic assessment system for computer users. It combines YOLOv11-based computer vision to assess body posture and workstation characteristics with brief self-report questionnaires covering ergonomic and psychosocial factors, including lighting, break habits, and perceived psychological strain. By integrating both assessments, the system generates ergonomic risk scores and targeted workplace improvement recommendations.”
1:40 - 2:20	Demo System	Kerangka pose tampil di atas video pengguna; saat leher menunduk, indikator berubah dan notifikasi muncul. Lanjut ke pengisian kuesioner, lalu dasbor hasil.	“Users capture their working posture using a camera, while YOLOv11 analyzes body keypoints and ergonomic indicators. They then complete a brief questionnaire on workstation conditions, work habits, and psychological strain. Both results are integrated into a dashboard displaying ergonomic risk scores and targeted improvement recommendations.”
2:20 - 2:40	Hasil	Grafik sederhana: perbandingan skor sistem vs. penilaian ahli, dan hubungan skor fisik–psikologis.	“ERGO-AI DESK aims to provide accessible and comprehensive ergonomic screening, with ongoing development and expert validation to improve assessment accuracy and support healthier, safer workplaces.”
2:40 - 3:00	Penutup	Presenter (zoom in) Logo, nama tim, afiliasi UGM, tagline	“ERGO-AI DESK integrates AI-based posture analysis and psychosocial assessment for multidomain ergonomic risk evaluation. Smarter assessment, healthier workplaces. Thank you.”